

**KAUA'I BEEF SUPPLY AND CONVERSION CAPACITY: TESTING MIDSTREAM
CONSTRAINTS AGAINST UPSTREAM EXPORT REALITIES**

Jennifer Andrade

BUS 620: Micro- and Macro-Economic Foundations for Managers

Individual Research Paper

October 6, 2026

KAUA'I BEEF SUPPLY AND CONVERSION CAPACITY: TESTING MIDSTREAM CONSTRAINTS AGAINST UPSTREAM EXPORT REALITIES

1. INTRODUCTION AND PROBLEM FRAMING

Kaua'i maintains 42,345 acres of pasture (HDOA, 2020) and 13,893 cattle and calves (USDA, 2022), but Hawai'i imports 85% to 90% of its food (Leung & Loke, 2008). Policy discourse frequently attributes this local beef deficit to downstream conversion bottlenecks—scarcity of slaughterhouse infrastructure, labor, and cold storage. Under this framing, maritime shipping disruptions are treated as exogenous shocks that cause shortages, rather than stress events that expose existing domestic conversion failures. This study evaluates whether downstream capacity truly acts as the primary constraint by benchmarking throughput against upstream herd allocations and market flows.

2. METHODOLOGY & FIELD EVIDENCE

Herd distributions and labor rates were established using the USDA 2022 Census of Agriculture and BLS OEWS May 2025 nonmetropolitan data. Primary midstream operational constraints were identified via audits of Kaua'i's main multi-user inspected processing facility (Operations manager, personal communications, October 1 and 2, 2026). According to the operations manager, other island facilities serve closed channels or remain inactive for regular commercial slaughter, positioning the audited plant as the primary gateway for independent producers.

3. EMPIRICAL FINDINGS: TESTING THE CAPACITY HYPOTHESES

3.1 Midstream Processing: Shallow Slack and Fixed Infrastructure Ceilings

Field evidence indicates the facility exhibits operational slack, but it is shallow and constrained by fixed infrastructure:

- Queue Friction & Utilization: Independent producers face ~1-week lead times.

Management reports slaughtering ~8 head on a Wednesday shift and a similarly sized weekend shift (~16 head per week). Operating frequency is governed strictly by inbound deliveries.

- **The Shallow Capacity Ceiling:** The facility's maximum throughput under existing infrastructure is ~25 head per week, leaving only ~9 head of unutilized slack. As illustrated in Figure 1, this represents a shallow midstream headroom that becomes a binding bottleneck at just ~13% retention of exported calves. Expanding beyond this ceiling is restricted by physical capital. Higher volume substantially increases wastewater volume and tank pumping, electricity for continuous refrigeration, and remote maintenance friction for equipment wear. Additionally, physical hanging rail space creates a hard capacity limit during peak weekend harvests.
- **Labor Structure:** Butchering runs part-time (3:00–6:00 AM) to accommodate staff's primary employment. Expanding to full-time requires competitive wages (BLS reports an \$18.90 median hourly wage for Hawai'i/Kaua'i nonmetro slaughterers, lagging the \$20.65+ resort baseline), which low fixed processing revenues cannot currently support.
- **Regulatory Friction:** USDA FSIS inspection is reliable and does not restrict harvest schedules.

3.2 The Upstream Structural Divergence: Comparative Advantage and Opportunity Cost

Under current market conditions, the upstream constraint binds first: the economic necessity of calf exports starves the facility of harvestable animals, leaving unutilized capacity of roughly 9 head per week. However, the midstream processing constraint forms a hard, secondary ceiling that binds almost immediately. If the island retained just 13 percent of its currently exported calves, the midstream labor and space bottleneck would become the primary binding constraint.

Kaua'i marketed 3,707 calves under 500 lbs and 1,911 cattle 500+ lbs (USDA, 2022). Calves represent ~66% of marketing, an annualized outflow of up to 71.3 calves per week (an upper bound including local transfers). The driver is comparative advantage: continental commercial feedlots leverage cheap, energy-dense grain rations, while Kaua'i's endowment favors extensive cow-calf pasture grazing. This creates a steep opportunity cost:

- Route A (Export): October 2026 continental auction prices for 400-500 lb steers hit \$4.70-\$5.47/lb (USDA AMS, 2026). At a conservative \$5.00/lb, a 450 lb calf grosses \$2,250. Deducting an assumed ~\$200 for freight, the immediate net cash settlement is ~\$2,050 at Year 0.
- Route B (Local Finishing): Retaining a calf to 1,000 lbs yields a 560 lb carcass. At \$4.75/lb wholesale (Operations manager, personal communication, October 1, 2026), gross revenue is \$2,660. Deducting \$350 for processing (Operations manager, personal communication, October 1, 2026) and an assumed \$1,200-\$1,400 for 24-month pasture carrying costs yields a nominal margin of \$910-\$1,110.
- Time and Risk Discounting: Discounting this nominal return over two years at 8% and factoring a 3% mortality risk ($1 / 1.08^2 \times 0.97 \approx 0.832$), the risk-adjusted Net Present Value (NPV) of local finishing is \$757 to \$923 per head.

The present-value deficit of local finishing is \$1,127 to \$1,293. Bridging this nominal gap at harvest requires an additional \$1,354-\$1,554 per head, or \$2.41-\$2.77/lb on a 560 lb carcass.

The estimated carrying cost of \$1,200 to \$1,400 per head accounts purely for direct inputs and holding costs over the finishing period. It does not include the opportunity cost of the pasture. In standard pasture management, holding a steer for grass-finishing displaces approximately 0.75 cow-calf pairs (USDA NRCS, 2003) that could otherwise generate continuous revenue through annual calf sales. Because this model excludes that foregone profit, it structurally underestimates the true economic gap between local finishing and mainland export.

4. POLICY IMPLICATIONS AND THE RESILIENCE EXTERNALITY

4.1 The Economic Objection to Production Subsidies

From a free-market efficiency perspective, subsidizing local pasture finishing is allocatively inefficient. Bridging the gap to retain one-third of exported calves (1,236 head) requires an annual public subsidy of ~\$1.79 million, or ~\$4.14 per retail pound. Critics argue it would be cheaper to purchase imported beef and maintain a frozen strategic county stockpile.

Dividing the proposed \$1.79 million annual state-funded price-bridge grant (calculated as 1,236 head multiplied by the \$1,450 nominal gap) by the 432,600 pounds of walking reserve beef yields an effective public subsidy of roughly \$4.14 per retail pound to secure this disaster capacity.

4.2 The Resilience Externality Defense

The \$1.79 million subsidy is a market distortion justified by the risk diversification it buys. First, Kauai's total weekly beef demand is approximately 113,000 pounds (calculated as the daily census of 73,000 residents and roughly 27,000 visitors consuming the national average of 59 pounds per capita annually; DBEDT, 2025; USDA ERS, 2024). Therefore, maintaining a 30-day (4.3-week) frozen reserve requires a stockpile of roughly 485,000 pounds of beef (113,000 lbs/week x 4.3 weeks). This demands steep capital warehousing outlays, high continuous electricity overhead, and inventory rotation spoilage. While a local processing plant also requires electricity, a "walking reserve" of live cattle holds its value during a prolonged hurricane grid failure, whereas a frozen stockpile spoils immediately. Phase 2 capital grants can equip the local facility with backup generation to ensure processing continues post-disaster.

The 1,236 retained head represent a "walking reserve." Assuming a 350-pound retail yield per head (derived from a 560-pound carcass at a 62.5% yield), this herd equates to approximately 432,600 pounds of retail beef. Against the island's normal baseline demand of roughly 113,000 pounds per week, this provides approximately 27 days of emergency supply. However, this walking reserve creates a severe processing paradox: during a maritime disruption, the reserve can only reach consumers as fast as the plant can harvest it. At the facility's hard ceiling of 25 head per week, it can process only 8,750 pounds of beef weekly—fulfilling less than 8 percent of island demand. While a frozen reserve can be distributed immediately, the utility of the walking reserve is fundamentally bottlenecked by the plant's midstream capacity, underscoring the critical need for Phase 2 expansion.

Second, domestic agriculture mitigates single-sector economic vulnerability. During Kaua'i's April 2020 pandemic lockdown, unemployment surged to 34.4% (Hawai'i DLIR, 2020).

Subsidizing this walking reserve provides vital non-tourism economic activity. This requires Phase 1 forward contracts to target institutional buyers (e.g., schools) whose budgets survive tourism crashes.

4.3 Sequential Policy Architecture and Coordination Problem

Policy must address a coordination problem: ranchers will not retain calves without slaughter access, while processors cannot justify lumpy capital investments without guaranteed volume. Interventions must deploy sequentially:

- Phase 1: Dynamic Contracts & Working Capital: The State must fund a dynamic price-bridge grant covering the \$2.41-\$2.77/lb premium, allowing schools to buy at wholesale (\$4.75/lb) while ranchers gross \$7.16-\$7.52/lb. Pegging this to a trailing 30-day AMS feeder price ensures the subsidy contracts if national cattle markets cool. In parallel, low-interest revolving credit lines must finance the 24-month grass-finishing working capital cycle.
- Phase 2: Immediate Infrastructure Grants: With a 25-head ceiling and 16-head current volume, slack is only ~9 head/week. Retaining just 13% of exported calves (~9 head) exhausts midstream capacity. Therefore, capital grants for wastewater, cooler rails, and backup generation must deploy simultaneously with Phase 1.

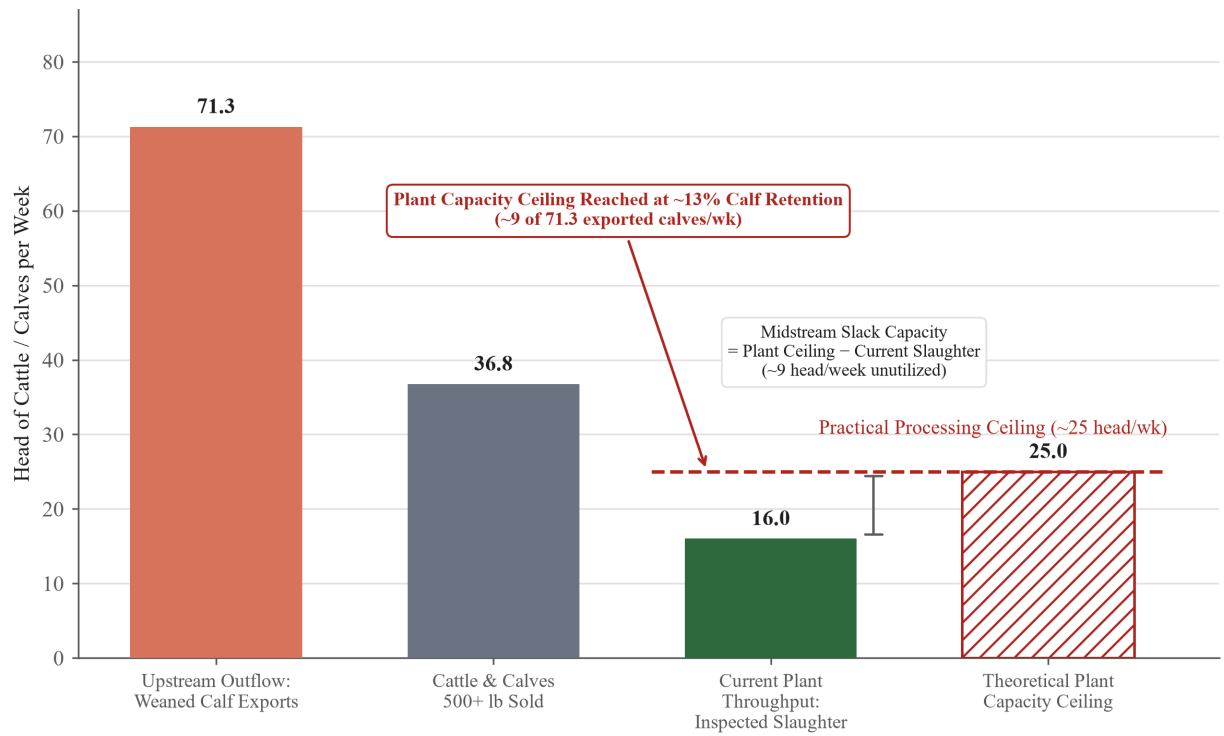
5. CONCLUSION

Kaua'i's beef deficit begins upstream due to the opportunity cost of calf exports. However, midstream infrastructure is bounded by a shallow 25-head weekly ceiling. Food security requires state-funded dynamic forward contracts to close the upstream margin gap, paired with immediate infrastructure grants to expand midstream capacity before retention volume exhausts the facility's limits.

References

- Bureau of Labor Statistics (BLS). (2025). May 2025 Metropolitan and Nonmetropolitan Area Occupational Employment and Wage Statistics: Hawaii / Kauai Nonmetropolitan Area (SOC 51-3023: Slaughterers and Meat Packers). U.S. Department of Labor.
- Hawai'i Department of Agriculture (HDOA). (2020). Statewide Agricultural Land Use Baseline 2020. State of Hawai'i.
- Hawai'i Department of Business, Economic Development and Tourism. (2025). Annual visitor research report and resident population estimates. State of Hawai'i.
- Hawai'i Department of Labor and Industrial Relations (DLIR). (2020). Local Area Unemployment Statistics (LAUS): Kaua'i County, April 2020. State of Hawai'i.
- Leung, P., & Loke, M. (2008). Economic Impacts of Increasing Hawai'i's Food Self-Sufficiency. Economic Issues (EI-16), CTAHR, University of Hawai'i at Manoa.
- U.S. Department of Agriculture (USDA) Agricultural Marketing Service (AMS). (2026). National Daily Feeder and Stocker Cattle AM Summary (Report SJ_LS850). USDA.
- U.S. Department of Agriculture (USDA) National Agricultural Statistics Service (NASS). (2022). 2022 Census of Agriculture, Volume 1, Chapter 2: County Level Data (Table 11: Cattle and Calves - Inventory and Sales: 2022 and 2017). USDA.
- U.S. Department of Agriculture, Economic Research Service. (2024). Food availability (per capita) data system.
- U.S. Department of Agriculture, Natural Resources Conservation Service. (2003). National range and pasture handbook.

Figure 1: Kaua'i Weekly Bovine Allocation vs. Inspected Processing Capacity



Sources: USDA NASS 2022 Census of Agriculture, Table 11 (annual sales ÷ 52); plant values from primary operational audit (personal communications, Oct. 1–2, 2026).